

An Aesthetic Cultural Relic Poster Generation Framework Based on Multi-target Learning and Multimodal Large Language Model

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Abstract

This paper presents CrePoster, a data-driven framework to generate aesthetic posters for Chinese cultural relics, aiming to enhance the exhibition experience and promote cultural spread. CrePoster comprises three modules: (i) object segmentation module, (ii) content generation module, and (iii) poster generation module. Upon processing a cultural relic image, the object segmentation module first leverages a cascaded U_2Net -SAM structure to obtain the visual target. Secondly, the content generation module utilizes a multi-target learning-enabled caption generator to produce professional captions. Thirdly, the Multimodal Large Language Model (MLLM) based poster generation module adaptively creates aesthetic parameters, including layout and color scheme, ultimately rendering them into refined posters.

CCS Concepts

• Applied computing → Media arts.

Keywords

Cultural relic, aesthetic poster, multi-target learning, social sharing

ACM Reference Format:

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1 Introduction

Cultural relics embody human wisdom and historical progress, carrying thought, culture, spirit, and emotion with high historical and artistic value. However, their ancient origins create barriers to

understanding their knowledge. These days, people enjoy sharing experiences on social media. However, in the museum scenario, a single image or long description decreases interest and limits cultural communication. Countering this issue, generating aesthetic cultural relic posters offers a promising solution to enhance visitor experience, deepen knowledge, and promote social sharing [11, 21].

With rapid AI advancements, poster generation has been explored for academic papers [12, 13], advertisements [5, 8, 17], and promotions [2, 19]. However, the cultural relic poster generation remains underexplored. Prior work mainly targets isolated issues like layout optimization [1, 4, 15], element matching [18], or background retrieval [6, 10]. These methods require pre-made textual information as input, making them ill-suited for cultural relics. Writing captions for cultural relics is challenging due to the required knowledge. Sheng et al. proposed an ancient Egyptian artifact caption framework [16], and Zheng et al. used a multi-classifier for cultural relic caption generation [20]. Yet, their slot-based models limit content richness, flexibility, and fluency.

To tackle these challenges, we propose CrePoster, an end-to-end framework that generates posters for Chinese cultural relics from a single image. CrePoster includes three modules (see Fig. 1): object segmentation module, content generation module, and poster generation module. First, a cascaded U^2Net -SAM network is employed for semantic segmentation to extract the key visual target. Then, a multi-target learning-enabled caption generator is utilized for caption generation. Last, an MLLM and multi-round optimization mechanism is developed to generate layout and color parameters, and ultimately renders an aesthetic poster.

2 Framework Design

2.1 Object Segmentation Module

Cultural relics possess surfaces with different textures, exhibiting different reflection states under the lights. Generally, images released by museums exhibit a high degree of fidelity, whereas photos taken by visitors during exhibitions may encounter issues related to lighting conditions, leading to potential challenges to the existing image segmentation approaches. CrePoster's object segmentation module utilizes a cascaded U_2Net -SAM [7, 14] network structure for precise extraction of the cultural relics' visual targets. Upon processing an input image, the U_2Net first identifies the coarse-grained outline. Subsequently, the pixel coordinates of the outline boundary

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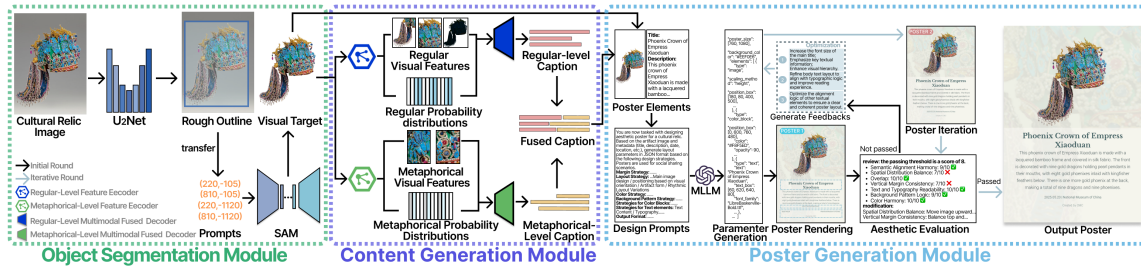


Figure 1: Framework of CrePoster, which automatically generates cultural relic posters.

are converted into prompts, which are utilized for guiding the SAM model in performing fine-grained image segmentation. Finally, a precise visual target of the cultural relic is obtained.

2.2 Content Generation Module

To precisely represent cultural relics' visual features, we propose a multi-target learning based caption generator, including a regular target codec (for visual features like color, shape, and pattern) and a metaphorical target codec (for semantic features). For example, in Chinese culture, plum blossom, orchid, bamboo, and chrysanthemum are known as the "Four Gentlemen," symbolizing noble character and moral integrity. Each codec employs a *VGGNet*-based encoder and an *LSTM*-based decoder to process the input visual targets. Multi-level features are encoded separately, fused, and then decoded into captions that reflect both visual attributes and cultural meanings. To train this module, we constructed the *MeArt* dataset, containing 8,310 high-quality cultural relic images and 20,515 expert-annotated captions [9, 20].

2.3 Poster Generation Module

We propose an automatic layout generation method driven by MLLM, enabling cross-modal semantic understanding and structured layout parameter output through two modules. In the first module, in collaboration with professional designers, we distilled three core design principles: visual hierarchy, focal point, and rhythm, to guide prompt construction. Poster parameters are then generated by an MLLM using the visual target and caption, and subsequently rendered into a poster image. The second module evaluates the generated posters across seven dimensions: (i) semantic alignment harmony, (ii) spatial distribution balance, (iii) overlap, (iv) vertical margin consistency, (v) text and typography readability, (vi) background pattern logic, and (vii) color harmony. Posters scoring below a defined threshold (8 in 10) are iteratively refined according to the modification suggestions generated by MLLM, until all criteria are satisfied for the final output.

2.4 Prototype Implementation

Given the widespread usage of online museums today, the prototype of CrePoster has been implemented on both web and mobile platforms to cater to users' online and offline experiences. The demonstrations of relevant interfaces are depicted in Fig.2. Upon clicking the "Upload" button, users can obtain aesthetic posters utilizing either Internet-sourced or real-time cultural relic images. Balancing generation speed with content diversity, the prototype system is configured to produce four distinct posters featuring

cultural relics per operation. In this way, it can satisfy both the immediate demand for poster generation in the museum scene and the creation demand of web-side cultural relic posters outside the exhibition scene.

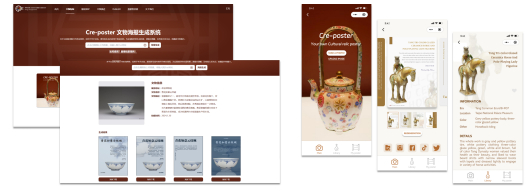


Figure 2: CrePoster prototypes on web and mobile platforms.

3 Demonstration

Examples of cultural relic posters generated by CrePoster are illustrated in Fig.3, showcasing that CrePoster is capable of (i) accurately extracting the crucial visual target of cultural relics, (ii) generating professional captions and detailed descriptive sentences, (iii) adaptively generating the appropriate aesthetic layout and visual harmony color schemes that reflect the cultural relic.



Figure 3: Examples of posters generated by CrePoster.

4 Conclusion

This paper introduces CrePoster, an automatic framework for Chinese cultural relic poster generation. Demonstrations show that the CrePoster can generate visually harmonious posters, enhancing visitor experience and promoting cultural knowledge dissemination. Future work may include extending CrePoster to support relics from diverse cultures and exploring multimedia-focused inspiration tools [3]. Moreover, incorporating continuous learning could enable personalized poster generation tailored to different user needs, such as museum visitors and professional designers.

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